EXP. NO. 8 DATE: 07/10/2025

IMAGE STEGANOGRAPHY

AIM

To hide a secret message within a cover image using a secret key such that others cannot detect the presence of the hidden message.

ALGORITHM

1. Start.
2. Input: Cover image, secret message, and secret key.
3. Convert the secret message into a text format.
4. Convert the secret message into its binary representation.
5. Converts the secret key into binary codes.
6. Set the bits-per-unit to zero.
7. Encode the secret message into the image pixels by XORing message bits with key bits.
8. Modify image pixel values accordingly.
9. Save the modified image as the stego image.
10. To extract the message, input the same key and reverse the process.
11. Output: Display the stego image and extract the hidden message when required.
12. End.

PROGRAM

```
import cv2
import string
import os

d = {}
c = {}

for i in range(255),
    d[chr(i)]=i
    c[i]= chr(i)

x=cv2.imread(r"C:\Users\TCS\Desktop\img.jpg")

i=x.shape[0]
j= x.shape[01]
print("Image dimensions:",i,j)

key = input("Enter key to edit (Security key):")
text= input("Enter text to hide:")

kl =0
tln= len(text)
z=0
n=0
m=0
l= len(text)

for i in range(l):
    x[n,m,z]=d[text[i]] ^ d[key[kl]]
    n= n+1
    m= (m+1) %3
    kl = (kl+ 1) % len(key)
```

```
cv2.imwrite("encrypted-img.jpg",x)
os.startfile("encrypted-img.jpg")
print("Data Hiding in Image completed
                          successfully")

kl=0
tlen=len(text)
z=0
n=0
m=0

ch= int(input("\n Enter 1 to extract data from
                                           Image:"))
if ch==1:
    key1=input("\n Re-enter key to extract text
    decrypt=""
    if key==key1:
        for i in range(l):
            decrypt+= c[x(n,m,z]^d[key[kl]]]
            n=n+1
            m=(m+1)%3
            kl=(kl+1)% len(key)
        print("\n Encrypted text was:", decrypt)
    else:
        print("key doesn't match")
else:
    print("Thank you. EXITING.")
```

OUTPUT:

Enter key to edit (Security key): secret123

Enter text to hide: hello all

Data Hiding in Image completed successfully.

Enter 1 to extract data from Image: 1

Re-enter key to extract text: secret123

Encrypted text was: hello all

RESULT

Thus, the secret message was successfully hidden in the image and later retrieved using the correct security key.